

Assignment – 1

Project: DOB to Personality Number



Problem Statement

Build a console application that:

- 1. Accepts the user's Date of Birth in numeric format (for example: 24/07/2001 or 24072001).**
- 2. Extracts all digits from the DOB.**
- 3. Adds the digits together.**
- 4. If the result has more than one digit, keep adding its digits again until a single digit is obtained.**
- 5. Display the personality type according to the mapping below.**

```

1  #include <iostream>
2  using namespace std;
3
4  int sum(int a) {
5      int pn = 0;
6      while(a > 0){
7          pn += a%10;
8          a/= 10;
9          if(pn>9){
10             pn = sum(pn);
11         }
12     }
13     return pn;
14 }
15
16 void pnlist(int pn){
17
18     switch(pn){
19         case 1:
20             cout << "The Leader";
21             break;
22         case 2:
23             cout << "The Peacemaker";
24             break;
25         case 3:
26             cout << "The Creative Communicator";
27             break;
28         case 4:
29             cout << "The Builder";
30             break;
31         case 5:
32             cout << "The Free Soul";
33             break;
34         case 6:
35             cout << "The Nurturer";
36             break;
37         case 7:
38             cout << "The Seeker";
39             break;
40         case 8:
41             cout << "The Powerhouse";
42             break;
43         case 9:
44             cout << "The Humanitarian";
45             break;
46         default:
47             cout << "ERROR";
48             break;
49     }
50 }
51
52
53 int main(){
54     int dob;
55     cout << "Enter your date of birth in DDMMYYYYY :";
56     cin >> dob;
57
58     cout << "you DOB : " << dob << endl;
59
60     int pn = sum(dob);
61
62     cout << "your Personality number is : " << pn << endl;
63
64     pnlist(pn);
65
66 }

```